Cobordism Distance

Knot Theory Research

October 29, 2026

Abstract

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1 Table

For $K, J \subset S^3$, We define the cobordism distance as $d_C(K, J) = g_4(K\# - J)$. This collapses to a metric in $\mathcal{C}$

$d_C(K,J)$	O	3_1	4_1	5_1	5_2
O	x	x	x	x	x
3_1	x	x	x	x	x
4_1	x	x	x	x	x
5_1	x	x	x	x	x
5_2	x	x	x	x	x

Table 1: Concordance distance $d_C(K, J)$ for prime knots up to 5 crossings.

1.1 Table

2 Identities

We have:

$$g_4(K) = g_4(K^r) \tag{1}$$

$$d(K, J) = g_4(\bar{K}, \bar{J}) \tag{2}$$

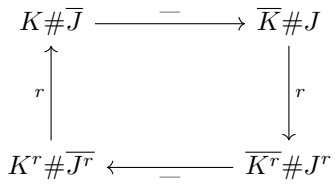
$$d(K^r, J^r) = d(K, J) \tag{3}$$

$$d(K^r, J) = d(K, J^r) \tag{4}$$

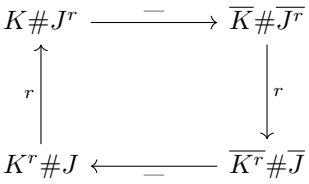
1. Equivalence Class: $K\#J$

$$\begin{array}{ccc} K\#J & \xrightarrow{\quad - \quad} & \overline{K}\#\overline{J} \\ {\scriptstyle r}\uparrow & & \downarrow{\scriptstyle r} \\ K^r\#J^r & \xleftarrow[\quad - \quad]{} & \overline{K^r}\#\overline{J^r} \end{array}$$

2. Equivalence Class: $K\#\overline{J}$



3. Equivalence Class: $K\#J^r$



4. Equivalence Class: $K\#\overline{J^r}$

